# AI Law Lab: Outside Data Access and Agent Web Reading

Which services the lab calls, when its agents go online, and how a site's wishes are honoured

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## Summary

The lab's language models run on its own hardware, and its experiments search its own versioned libraries. It reaches beyond the university in three situations only, each allowed by a person: a researcher searching the public legal databases or adding web pages to a library on the **Legal Sources** page; a researcher drafting a role-play cast from a news story or opinion given by its address; and a research run (the **agentic workflow**) whose researcher ticked legal databases or the open web. Role-play and document-analysis runs never go online.

What leaves the server is a search term or an address, never a document a researcher uploaded. What comes back is saved into a library before anything cites it, so every citation names a fixed copy: a document, a library and a version, with the address it came from and when it was read. And since October 2026 a page is read on the lab's own initiative -- by a research agent, or by a crawl -- only if the site has not said it does not want AI systems reading it (section 5).

## 1 · Everything the server connects to

| Destination | What is sent | When |
|---|---|---|
| The two DGX Sparks (Tailscale, inside the university) | Prompts and text to embed | Every model call; never leaves university hardware |
| CourtListener, Federal Register, eCFR, govinfo, SEC EDGAR | A search term, or the identifier of a result | A person searches or adds on Legal Sources; an agent searches or reads (databases allowed) |
| Brave Search API | A search query written by an agent | An agent searches the open web (open web allowed) |
| Any public website | A request for one address | An agent reads a page; a page named in the brief; a crawl; a person adds or re-checks a link, or drafts a cast from one |
| UW Shibboleth, InCommon (mdq.incommon.org), Google, Microsoft | Sign-in exchanges (Keycloak) | Someone signs in with that provider |

No prompt, document or answer is sent to a commercial model API. The search terms an agent writes do leave the university (to the database or to Brave), and they are drawn from the research brief: a brief that must stay confidential should be run with both outside permissions off.

## 2 · The legal databases

Five public repositories are reached through the interfaces their publishers provide for programs to use (`sources.py`). Each is a staging step, not a live oracle: a search returns candidates, and nothing is relied on until it has been fetched, chunked, embedded and saved into a library, with its canonical address and retrieval time in the document's metadata.

| Database | Interface and credential | What it holds; notes |
|---|---|---|
| CourtListener | REST API v4, courtlistener.com/api/rest/v4; API token (`AILAWLAB_COURTLISTENER_TOKEN`, set on the server) | Opinions from federal and state courts (Free Law Project). Without a token only the search snippet can be saved, and it is labelled as a snippet. |
| Federal Register | API v1, federalregister.gov/api/v1; no key | Proposed and final rules, notices, presidential documents since 1994; full text where published, else the abstract, flagged as such. |
| eCFR | ecfr.gov/api (search and versioner); no key | The Code of Federal Regulations as in force, read as of each title's latest issue date. |
| govinfo (GPO) | api.govinfo.gov; api.data.gov key (`AILAWLAB_GOVINFO_API_KEY`, set on the server) | US Code, Public Laws, bills, reports; HTML or text rendition, else the PDF. |
| SEC EDGAR | efts.sec.gov full-text search; www.sec.gov/Archives; no key | Company filings since 2001. Requests name the lab and a contact address, as the SEC's fair-access policy asks. |

Credentials live in the server's `.env`, never in the code or the browser. A database whose credential is missing reports itself unavailable rather than failing a page.

**Pacing.** A person's search goes straight through. Research agents call far more often, several at a time, so each database's requests from agents are spaced at least one second apart across every run on the server; a database that answers 429 ("too many requests") is left alone for three minutes, and agents are told to search elsewhere meanwhile (`network_tools.paced`). One agent search asks each database for four results.

**Open-web search.** Brave's Search API (`AILAWLAB_BRAVE_API_KEY`, set on the server) answers an agent's web searches with eight results each. Calls are spaced 1.1 seconds apart, the API's per-second limit, and one 429 is retried after two seconds. Results are leads, numbered [W1], [W2]...: an agent cannot cite a search result, only what it read.

**Why robots.txt does not govern these.** robots.txt addresses crawlers wandering a website. These interfaces exist so that programs can query them, and their use is governed by their own terms, keys and rate limits, which the lab follows. The documents they return are largely government works or public filings; that a document is public does not settle every use of it, and researchers remain responsible for theirs.

## 3 · When a research run goes online

An agentic-workflow run has two permissions, each the researcher's to grant on the plan before the run starts: search the legal databases and search the open web (which includes reading any public page). With both off, the run is entirely local. With either on, network access happens at these points and no others:

- **Planning: none.** The planner drafts sub-questions from the brief on the Sparks. It notes any web addresses written in the brief, but does not open them.
- **Before research: pages named in the brief.** If the open web is allowed, each address in the brief is read once, before any agent starts, so every agent has it. These reads pass the same checks as an agent's (section 5).
- **During research: the agents' own tools.** Each research agent (up to four at once) decides for itself when to call `search_databases`, `search_web` and `read`, as its sub-question needs. Reading is not budgeted; the run's time limit (two hours by default) and the researcher's Stop button end it. A document is read once per run, however many agents ask for it.
- **Write-up: none.** The lead agent writes the answer from the findings and the passages already read.

Everything read is saved into a library -- by default "Fetched: <experiment name>" -- which records a new version, and the run is pinned to it, so later library searches by any agent cover the new document too. The run page then lists every search with the results it gave, every document read (with the library version it was saved to), and every page that was not read because its site does not allow it.

## 4 · What a person adds

On the **Legal Sources** page a person can search the databases and add chosen results; add web pages by address (up to 20 at a time) and later check them again, which re-reads each page and rebuilds its passages only if the text changed (`web_links.py`); or crawl a page's links into a library (`crawler.py`). In the cast builder a person can draft a role-play from a page given by address (`source_material.py`). Adding a link or drafting from one is that person reading the page, as they would in a browser, so it is not put through the AI checks below; a crawl, which reads many pages automatically, is.

## 5 · Does the site let AI agents read it?

Before a research agent's read and before every page of a crawl, the lab asks what the site has published about automated and AI use, and obeys it (`site_rules.py`). Figure 1 shows the order for an agent.

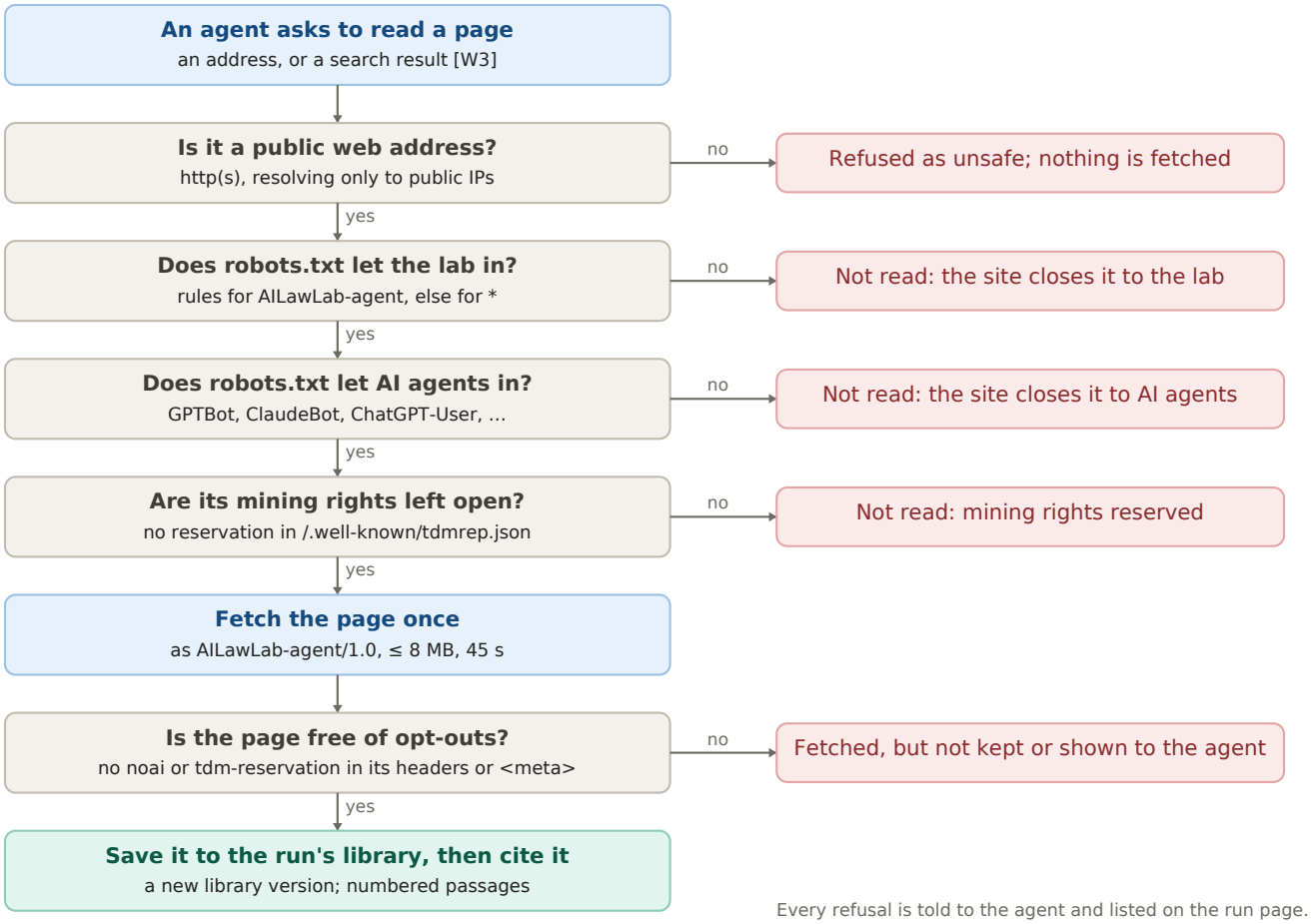



Figure 1. How a research agent's request to read a page is decided. A crawl asks the same questions under its own name, AILawLab-crawler, and records a skipped page as "not open to AI use".

- **robots.txt, for the lab** (RFC 9309). Read first. Agents are known to it as `AILawLab-agent`, crawls as `AILawLab-crawler`; a site may write rules for either, or for all crawlers (`*`). The most specific matching rule decides, and the `*` and `$` wildcards are understood -- Python's built-in reader ignores them, which would have let `Disallow: /*.pdf$` through. A missing robots.txt sets no rules; one that cannot be read because the site is failing means nothing is read.
- **robots.txt, for AI agents.** Many sites now turn named AI agents away while admitting search engines. If the site's robots.txt closes this page to any of ChatGPT-User, Claude-User, Perplexity-User, MistralAI-User, GPTBot, OAI-SearchBot, ClaudeBot, Claude-SearchBot, anthropic-ai, PerplexityBot, CCBot, Google-Extended, Applebot-Extended, Bytespider, meta-externalagent or cohere-ai, the lab treats that as a refusal of AI systems in general, which it is one of. Only rules written for those agents by name count.
- **Text-and-data-mining reservations** (W3C TDM Reservation Protocol). A site can reserve its TDM rights in `/.well-known/tdmrep.json`, by location; on the page in a `tdm-reservation: 1` header; or in a `<meta name="tdm-reservation">` tag. A reservation means not read; its licensing address, if given, is passed on.
- **"noai" on the page.** A `noai` directive in an `X-Robots-Tag` header (for every agent, or for the lab's name) or in a robots `<meta>` tag in the page's head. These can only be seen once the page arrives, so such a page is fetched but neither kept nor shown to the agent.

A refused agent is told plainly why and asked to find another source; the trace records it; and the run page lists each refused address with its reason. A site's rules are read at most once an hour per site, across all runs (RFC 9309 allows a day), and a failing site is asked again next time rather than remembered. If a page redirects to another site, that site's rules are checked too.

## 6 · Other safeguards on every fetch

- **Public addresses only.** This server can reach the Sparks, its database and the university network, so a link is fetched only if it is `http`/`https` and its host resolves exclusively to public addresses; every redirect (at most five) is checked again (`source_material.check_public_url`).
- **Bounded.** A response is cut off at 8 MB and 45 seconds; a page kept in a library keeps up to 100,000 words, a cast source 12,000.
- **Named.** Requests say who is asking: `AILawLab-agent/1.0` for agents, `AILawLab-crawler/1.0` for crawls, each with the lab's address and a contact, so a site owner can see who is reading and how to reach us.
- **Polite crawling.** A crawl stays on one website and goes one level deep, fetches one page at a time at least two seconds apart (longer if robots.txt asks, up to 30), takes at most 100 pages, stops on a 429 or 503 or three failures in a row, runs one at a time per site, and stores the licence or copyright notice it finds with every document.
- **Recorded.** Searches with their results, documents read with their library versions, pages refused with the reason, and every crawl with the rules it ran under and the outcome for each link, are kept in the database.

## 7 · The lab's own interface

The browser talks to the lab through JSON endpoints under `/ai_law_lab/api/`. They are the portal's plumbing, not a public API: every one needs a signed-in, approved account, and changes need the researcher or admin role (`web/auth.py`). The ones that reach outside are `/api/sources/search` and `/api/sources/ingest` (databases), `/api/sources/links` and the two `refresh` endpoints (web links), `/api/crawls` and `/api/crawls/preview` (crawls), and `/api/source/read` (a cast source). Runs are launched from the experiment page and followed live through `/api/runs/{id}/stream`.

## 8 · Limits and open questions

- The AI-agent list is maintained by hand. New agents appear; the list is in `site_rules.AI_AGENTS` and should be reviewed each term.
- Machine-readable AI preferences are still being standardized (the IETF AI Preferences working group). The lab follows robots.txt, `noai` and TDMRep, which sites use today, and should adopt the IETF vocabulary once it settles.
- A site's permission to read is not a licence to reuse. The licence notice a crawl records, and the address and retrieval time kept with every document, are there so a researcher can check before relying on, quoting or publishing what was collected.
- Pages a person adds by hand are not put through the AI checks (section 4). If the lab's practice should be stricter, the same check can be applied there with a line of code.

## Where it lives

| Topic | Source |
|---|---|
| Legal databases | `sources.py` |
| Agents' searching and reading; pacing | `network_tools.py, tools.py` |
| Run permissions and pages named in the brief | `research.py, experiments.py` |
| Site permission checks | `site_rules.py` |
| Fetching, address guard, limits | `source_material.py` |
| Web links in a library | `web_links.py` |
| Crawls | `crawler.py` |
| Settings quoted here | `config.py (each overridable by an AILAWLAB_ variable)` |
| Run page: read online, refused | `web/views.py, web/templates/run.html` |

Paths under `src/ailawlab/`. Tests: `tests/test_site_rules.py`, `tests/test_network_tools.py`, `tests/test_crawler.py`. Rebuild this note with `.venv/bin/python docs/network-access/build.py`.